

VR AI NOTES

9-Week Project Timeline

Week 1: Foundation & API Integration

02-13-2026

- ☒ ~~Unity project setup~~
 - ☒ ~~XR simulator~~
 - ☒ ~~Basic scene with heart models~~
 - ☒ ~~Set up Groq API (LLM, STT, TTS)~~
 - ☒ ~~Test full pipeline in desktop mode~~
 - ☒ ~~Initial VR build & headset test~~
 - ☐ **Milestone:** AI conversation working, tested in VR
-

Week 2: VR Interaction & Context System

02-20-2026

- ☒ ~~Build and test on Quest headset regularly~~
 - ☒ ~~Implement gaze detection (what user is looking at)~~
 - ☒ ~~Create VisualizationData scriptable object system~~
 - ☒ ~~Add metadata to heart models~~
 - ☐ Update LLM system prompt to be generic (not heart-specific)
 - ☒ ~~Test context-aware responses in VR~~
 - ☒ **Milestone:** Bot knows what you're looking at and responds contextually
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Week 4: Bot Model & Basic Animation

03-06-2026

- ☒ Source/import humanoid bot model
 - ☒ Position in scene, test scale in VR
 - ☒ Add basic animations (idle, talking gesture)
 - ☐ Test bot visibility and presence in VR
 - ☒ Adjust positioning and scale based on VR comfort
 - ☐ **Milestone:** Bot model present and animated in VR
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Week 5: Bot Integration & Audio

03-13-2026

- ☒ Connect bot animations to audio playback
 - ☒ Add simple lip sync or jaw movement
 - ☒ Implement bot gaze (looks at user or at visualization)
 - ☒ Test audio spatialization in VR
 - ☒ Refine bot behavior based on VR testing
 - ☐ **Milestone:** Bot speaks with synchronized animation
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Week 5 & 6: Particle System

04-03-2026

- ☒ Design particle swarm
 - ☒ Implement responsive particle behavior
 - ☒ Particles react to AI state (idle, listening, speaking)
 - ☒ Position and tune for VR visibility
 - ☐ **Milestone:** Both embodiments functional in VR
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Week 7: Scene Polish & Lighting

04-10-2026

- ☒ Finalize scene layout based on VR feedback
 - ☒ Implement proper lighting
 - ☒ Add environment details (pedestals, subtle UI)
 - ☐ Performance optimization for Quest (maintain fps)
 - ☒ Test with different users for comfort
 - ☐ **Milestone:** Polished, well-lit, comfortable VR scene
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Week 8: Final Polish & Documentation

04-24-2026

- Final testing with multiple visualization types
 - Record demo video (both embodiments)
 - Update documentation
 - **Milestone:** Complete, demo-ready, documented
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10/02/2026

- Scene
 - Basic scene with human heart. Could ask the bot to tell how two heart models differ or maybe which cell type occurs the most?

- Use HRA 2.4 models for it.

<https://github.com/cns-iu/hra-organ-gallery-in-vr-development/tree/main/hra-organ-gallery/Assets/Models/Organs/2.4/HiRes>

- No constraints on scene wrt the skybox, env, etc. There's flexibility.

- Bot

Humanoid

- Humanoid model - stay away from the uncanny valley.

- Look at Fab, Unity assets, Sketcfab.

- Could import animations from Mixamo, the model needs to be properly rigged for this.

- If not mixamo, use Blender to create animations.

Particles

- Use a particle system. Something like Navi from Legend of Zelda

<https://youtube.com/shorts/ShXbpyPgbOg?si=ExUsNCbHkE6o5rOU>

- An entity that follows you around (not in an annoying way)

- Maybe a pulsing swarm instead of something that fades over time so that the user does not feel like they have limited time to interact with the entity.

- Andi mentioned HRA red fading to white (color over lifetime)

12/02/2026

Pricing:

Complete AI Pipeline:

1. **STT (Speech-to-Text):** User talks in VR → converts to text
2. **LLM (Language Model):** Text question → generates text answer
3. **TTS (Text-to-Speech):** Text answer → audio for bot to speak

Option A: OpenAI Ecosystem (simplest)

- **STT:** OpenAI Whisper API - \$0.006 per minute

- **LLM:** GPT-4o mini - very cheap (~\$0.15 per 1M input tokens)
- **TTS:** OpenAI TTS - \$15 per 1M characters
- **Pros:** Single API provider, great documentation, reliable
- **Estimated cost for 7-8 weeks:** \$10-30 total

Option B: Mix & Match (cheapest)

- **STT:** OpenAI Whisper - \$0.006/min
- **LLM:** GroqAI (LLaMa 3) - FREE tier is generous, super fast
- **TTS:** ElevenLabs free tier or \$5/month
- **Pros:** Cheapest, GroqAI is extremely fast
- **Estimated cost:** \$0-15 total
- **Cons:** Managing multiple APIs

ElevenLabs has a built-in Unity SDK, and is explicitly marketed for VR.

13/02/2026

- Will have to look into lighting. The models need to be well lit from all angles.
- Create a timeline for the project. We have 9 weeks. April 24th.
- The bot can have basic animations rn (idle and talk)
- This could be a scene in HRA. The idea is for the user to be able to have a conversation about the visualizations and models, and have their queries answered in real time.

17/02/2026

- Groq API key:
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

Orpheus (via Groq Cloud) is a fast, high-quality open-source text-to-speech model. It is designed for real-time applications, often achieving under 200ms time-to-first-byte when powered by Groq's specialized Language Processing Units (LPUs). **ElevenLabs** is a proprietary platform known for its incredibly natural, emotionally expressive voices, and extensive voice cloning capabilities. [↗](#)

At-a-Glance Comparison

Feature ↗	Orpheus (Groq)	ElevenLabs
Primary Strength	Ultra-low latency (Speed)	High-fidelity voice/emotion
Speed	Sub-200ms, specialized for RT	Very fast, but typically higher latency
Model Type	Open-source (Canopy Labs)	Proprietary
Voice Quality	Very high, natural (Arabic/English)	Industry-leading, "too perfect"
Best For	Real-time agents, low-cost prod	Content creation, high-end cloning

Key Differences

1. Latency and Speed (Groq Advantage)

Groq's infrastructure is designed specifically to reduce latency for AI models. When using Orpheus, the "time-to-first-byte" can be under 200ms, which is critical for natural, interactive voice conversations. This makes it a strong contender for "real-time" voice agents. [↗](#)

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- Groq has all 3 services (STT, LLM, TTS)
 - STT: Whisper Large v3 Turbo
 - LLM: Llama 4 Scout
 - TTS: Orpheus EnglishWill try using Orpheus (has fewer voices but very low latency)
- Context for bot:
 - Scriptable Objects that can be attached to each visualization. Can use JSON/text files that provide metadata from external files (useful for non-programmers but SO better imo)

- The bot can look at a game object and detect through raycast and then pull up context about that specific game object. Makes it more dynamic that way.

19/02/2026

- Used up 3.5k tokens for orpheus tts. The limit is 3.6k. That makes it expensive.
- Elevenlabs Flash on the other hand will give me about 20 minutes of audio where one minute of audio costs approximately 1000 characters.

Compare plans						
< Text to Speech Speech to Text Agents Voice Changer Sound Effects Voice Isolator Music Image & Video Auton >						
	Free	Starter	Creator	Pro	Scale	Business
Multilingual v2/v3						
Minutes included ⓘ	~10	~30	~100	~500	~2,000	~11,000
Extra minutes ⓘ	×	×	~\$0.30/minute	~\$0.24/minute	~\$0.18/minute	~\$0.12/minute
Audio quality ⓘ	128 kbps, 44.1kHz	128 kbps, 44.1kHz	128 & 192 kbps (via API), 44.1kHz	128 & 192 kbps (via Studio & API), 44.1kHz	128 & 192 kbps (via Studio & API), 44.1kHz	128 & 192 kbps (via Studio & API), 44.1kHz
Flash						
Minutes included ⓘ	~20	~60	~200	~1,000	~4,000	~22,000
Extra minutes ⓘ	×	×	~\$0.15/minute	~\$0.12/minute	~\$0.09/minute	~\$0.06/minute
Audio quality ⓘ	128 kbps, 44.1kHz	128 kbps, 44.1kHz	128 & 192 kbps (via API), 44.1kHz	128 & 192 kbps (via Studio & API), 44.1kHz	128 & 192 kbps (via Studio & API), 44.1kHz	128 & 192 kbps (via Studio & API), 44.1kHz

- The starter tier is \$5 month and increases usage limit quite a bit.

20/02/2026

- ElevenLabs API key:

Sk_c4fe0a87504049908eb3272b634071d92551b29621d3c450

- Super low latency: ~75ms

- 40k character limit (Orpheus has 3.6k). Could use the free tier for now and see if we need the \$5/month in the future.

24/02/2026

- Try multiple visualization types in the scene and see how that compares to the 3d models and how the AI Manager interacts with those as well.
- Updated Unity version. Using Unity 6.3 LTS (6000.9f1) now.
- Updates some of the deprecated assets and components that I used in the 2022 version.
- Using XR Interaction Simulator instead of XR Device Simulator now since that's Legacy and lacks many of the newer features.

26/02/2026

- Downloaded free 3d models from SketchFab (.blend or .fbx). Not sure if they're rigged.

Updates for today (March 06, 2026):

Still working on the particle swarm. Yesterday I spent some time going through YouTube tutorials on Blender animations and general tools to get more comfortable with the working. The animations are taking a bit longer than expected, but I'm making progress and have a clearer idea of what I need to do. Will keep pushing on it today.

10/03/2026

- **Type 2 — Orb/Sphere with implied structure.** The white smoky sphere and the blue cellular/energy orb both have a **loosely spherical form** that feels contained but turbulent inside. The blue one especially is interesting because it actually looks like a **cell** — which is a fantastic happy accident given your anatomical context. It has that organic membrane quality with an energetic core.

- This direction has a clear **center of gravity** — the entity "lives" in a defined space, which makes it easier for users to know where to look and talk to. It feels more like a *being* while still being abstract.
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- **My honest take combining both types:**
- I'd actually suggest a **hybrid** — a loose orb as the resting/idle state that occasionally trails off into ribbons or wisps when active or speaking. The sphere gives it a home base, the flowing tendrils give it expressiveness.
- The blue cellular orb is particularly compelling for your project because it visually rhymes with the anatomical content users are exploring.

So the concept would be something like:

Idle state — a gently pulsing orb, particles slowly orbiting/churning within the sphere boundary, calm blue-purple glow

Listening state — orb tightens slightly, maybe a subtle inward pull, indicating attention

Speaking/active state — tendrils start trailing off from the orb, ribbons of smoke/energy flowing outward, more chaotic and expressive, brighter

Thinking/processing — could do a fast swirl or a brief contraction before expanding again

This gives users clear **readable states** which is important for UX in VR — they need to know when to talk and when to wait.

When adding to GITHUB add to ARCHIVES folder

Week 1 | Feb 13, 2026 - Foundation & AI Pipeline

- Set up Unity project targeting Meta Quest (Android, ARM64)
- Built basic scene with HRA 2.4 heart models
- Implemented full AI pipeline: Groq Whisper (STT), Llama (LLM), ElevenLabs eleven_flash_v2_5 (TTS). Eventually switched from Groq Orpheus TTS early on due to very restrictive rate limits
- Created core scripts: AIManager, SpeechToText, LLM, TextToSpeech, BotController
- Tested full pipeline in desktop simulator and on headset - voice input and spoken responses working

Week 2 | Feb 20, 2026 - VR Interaction & Context System

- Switched controller input from trigger to grip for voice recording
- Implemented gaze detection via raycasting — detects which model the user is looking at and injects relevant metadata into the LLM prompt
- Built VisualizationData ScriptableObject system for per-model context — generalized to work with any tagged object, not just hearts

Week 3 | Feb 24, 2026 - Unity version update

- Migrated from Unity 2022.3 to Unity 6 (6000.3.09f1), URP
- Rebuilt XR Origin from scratch using Unity 6 prefabs rather than patching deprecated components
- Updated build settings for Quest: Render Graph, Android API 29+, ARM64, internet access for AI pipeline
- Confirmed build runs on Quest hardware

Week 4 | Mar 2–6, 2026 - Robot & Particle Swarm Started

- Sourced and tested multiple free robot models from Fab, Unity Asset Store, and SketchFab - most had rigging issues or weren't in T-pose, making Mixamo auto-rigging and animation mapping difficult
- One model from Fab was well-rigged and worked initially with idle and talk animations
- Started particle swarm work - evaluated Blender vs Unity VFX Graph, decided on VFX Graph since baked Blender animations can't respond to AI states at runtime

Week 5 | Mar 9–13, 2026 - Particle Swarm Build / Robot Issues

- Built particle swarm in VFX Graph: sphere emitter, turbulence, Point Attractor, Additive blend mode
- Created glowing orb core with HDR emission material, added Bloom post-processing via Global Volume
- Wrote CorePulse.cs to animate emission intensity via sine wave
- Started OrbitalSwirlVFX layer with Sphere emitter - particle drift issue from turbulence
- Robot model that had been working started distorting/stretching during testing - spent time debugging, couldn't resolve, scrapped the model
- Currently evaluating options for a replacement - free models have been exhausted, looking into purchasing a properly rigged model

Week 6 | Mar 21–26, 2026 — Particle Swarm State System & Build Testing

- Implemented AI state-driven behavior: idle, listening, and speaking states drive particle behavior and core color/emission in real time via SwarmStateManager
- Tested particle swarm build on Quest — VFX effects less visible on device than in editor, particle drift more noticeable in VR, continuing to tune
- Discussing robot model options with Andi — plan to purchase a properly rigged model to move forward

31st March

<https://www.fab.com/listings/3d6a744d-of70-4fb9-9b4c-059fead98ee8>

This one is available on Fab and Unity Asset Store. Has rigged animations. In the preview, it shows a mixamo animation which would mean that the model would work on mixamo auto-rig.

<https://www.fab.com/listings/81899db6-0a6d-4515-b795-9b6ac8758ad6>

Fully rigged and mixamo compatible as well.

02 April, 2026

-Made changes to tts and AIManager. The bot now stops talking when it starts listening so interruption is easier.

03 April, 2026

- Imported bots. One of them is creepy so I scrapped that.
- Added textures and materials
- Added animations to the bot. Looping talk for now.
- Tested in VR for the scale

07 April, 2026

NEED TO DO A LITERATURE REVIEW

Basically just go through what papers might already exist with VR-AI integration (it is likely they may have robots but not a particle system embodiment)

Just compare these and talk about how some things work and why, or why some things do not work or are limited in scope.

- Might need to buy ElevenLabs for a month? Characters and token limits are monthly. So we might run out of tokens for the demo.
- Assets had Built In shaders and we're using URP so converted using Render Pipeline Converter

14 April, 2026

- Add lighting for the swarm
- For literature review, include the whole scriptable object workflow.
- Added more data from UBERON ids, from here

https://www.ebi.ac.uk/ols4/ontologies/uberon/classes/http%3A%2F%2Fpurl.obolibrary.org%2Fobo%2FUBERON_0000948

April 16, 2026

- Added lights for heart models
- Fixed Gaze detection raycast issues. Changed raycast to spherical so it detects models better.
- Swarm follows you now.
- Swarm while talking changes the color of the directional light so that the swarm stands out clearly.
-